

Tide retrospective: the all-order expansion

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

The previous tide formalised the first correction of the quartic-perturbed Gaussian partition function and recovered the coefficient from it. The natural completion of the germbij Section 7.4 ladder for this potential is the *whole* expansion: every coefficient explicit, an elementary remainder at every order. The deliberation settled the route (a fundamental-theorem induction for the exponential Taylor remainder, over both the Mathlib Taylor API and alternating-series arguments) and independently verified the coefficient bookkeeping before any Lean was written.

2 The thing formalised

Six declarations in `Laplace/OneD/RecoveryAllOrder.lean`, sorry-free with zero warnings. The exponential Taylor remainder $E_n(s) = e^{-s} - \sum_{j < n} (-s)^j / j!$ with its bound¹ $|E_n(s)| \leq s^n / n!$ for all n and all $s \geq 0$, and the expansion²: for $b \geq 0$, $t > 0$, and every order n ,

$$\left| Z_b(t) - \sqrt{2\pi} \sum_{j=0}^n \frac{(-b)^j (4j-1)!!}{j!} t^{-(j+1/2)} \right| \leq \sqrt{2\pi} \frac{b^{n+1} (4n+3)!!}{(n+1)!} t^{-(n+3/2)}.$$

The $n = 1$ case subsumes the previous tide's bound with the sharp constant $105/2$. The numerical check at $n = 2$ (executed, archived in the tide log) shows the bound holding and nearly saturating at large t , as the alternating structure predicts.

3 Proof strategy

For the remainder: $E_0 = e^{-s} \in (0, 1]$; $E_{n+1}(0) = 0$ and $E'_{n+1} = -E_n$, so $E_{n+1}(s) = -\int_0^s E_n$, and the inductive bound integrates to $s^{n+1} / (n+1)!$. Interval integration, integral monotonicity, and the polynomial antiderivative are the only ingredients. For the expansion: split the integrand into the Gaussian factor times the expanded quartic exponential; each of the $n+1$ terms integrates exactly against the scale- t moment $(4j-1)!! \sqrt{2\pi} t^{-(2j+1/2)}$, and the prefactor $(-tb)^j$ converts the exponent to $t^{-(j+1/2)}$ through one isolated `rpow` helper; the remainder term is dominated by the $(n+1)$ -st moment through the pointwise bound. As in the first-order tide, the cancellation is exact: the expansion error *is* the integrated remainder.

¹`abs_expRemainder.le`.

²`quartic_partition_expansion_allOrder`.

4 Roadblocks and resolutions

Six build iterations, every one elaboration-level; the mathematics never moved. New entries for the gotcha catalogue: the `HasDerivAt.sub/sum` combinators produce `Pi`-form functions (a derivative-side sibling of the documented `Pi.add` integral gotcha), bridged by a `funext` identity; the interval-integral norm bound needs its measure passed explicitly when the `have` has no expected type; the double-factorial notation is scoped (`open scoped Nat`); and an equality `calc` against a $\leq$ goal belongs under `le_of_eq` from the start.

5 What was Mathlib, what was new

Mathlib: the fundamental theorem of calculus for interval integrals, integral monotonicity, the polynomial antiderivative³, derivative combinators, and integral linearity over finite sums. Seabed: the scale- t moments and monomial integrability, consumed now at every order simultaneously. New: the remainder recursion packaging and the observation that the whole ladder needs no analysis beyond it.

6 Lessons

The `Pi`-form mismatch family now has three members (integral `add`, ratio `div`, derivative `sub/sum`); the uniform repair is a type-ascribed or `funext` bridge to single-lambda form, and it is worth reaching for immediately whenever a combinator-built hypothesis must match a goal syntactically. Isolating `rpow` algebra in one helper (the consult's advice) kept all six iterations away from the exponent arithmetic.

7 Follow-ups

- The deferred coefficient-limit packaging (each coefficient as the limit of the rescaled remainder), now with a stable expansion API to build on.
- The same ladder for a general even-monomial perturbation $x^{2k} + b x^{2m}$: the proof is potential-generic except for the moment indices.
- Tag the note's Section 7.4 dimension-one item with the all-order theorem once merged.

³`integral.pow`.